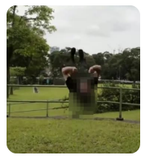
AutoMimic harness — programs measure, a foundation model decides, rules guard; runs (b), owns the memory (c), repairs the data (a)

tools: eval · critic · render | optional human feedback in words → **LLM decider** | guard: reward hacking | guard: G1 EDU limits · rollback | memory: library + decisions

(a) Internet video → reference



Internet video
monocular, in the wild
(Bilibili backflip)



Human recon.
GVHMR → SMPL
+ scene mesh (ext.)

VLM annotation
phases · contacts · take-off / landing





Retarget to G1
GMR · foot lock
ground fix

Reference motion q*(t)
+ per-frame phase labels

(b) Recursive self-improvement loop (per skill)

① Train
RL motion tracking (PPO)
4096 parallel envs, MuJoCo-Warp
rewards · terminations · start-state prior
real-robot limits (G1 EDU, knee 120 N·m)
warm start from the best checkpoint
≈1 h / round · one laptop GPU (8 GB)

policy π_k

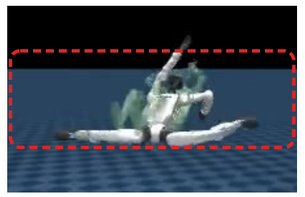
② Evaluate
full rollout from standing, every frame
robot (white) vs. reference ghost (green)



per-step metrics from the library

video +
per-step table

③ Critic: measure
library checks + uncovered deviations
joint · body · jitter · symmetry · contact · timing



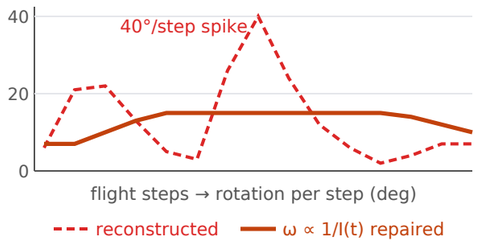
flight, step 101
legs 1.26 m apart
hips +2.1 rad
(ref. tuck −1.57)
→ new entry

round 4 of the backflip, real rollout

crouch · take-off · flight · landing · stand

issues + frames

④ LLM decide & repair
ranks failures (video + numbers)
template → hand-written code → ref. repair
scene: box height · wall pose (ext.)
rules only suggest · decision logged



edits
→ round k+1

repaired reference (dashed)

reference + labels

round k
until stable
decision log

(c) Diagnostic library (3 levels)

checks

failure signature	learned fix
L0 generic · any motion	
reward gated off at a phase switch	gate fix
fix std ≪ error (no gradient)	std ≈ error
fails at 20 ms latency	delay training
L1 family · aerial skills	
legs spread in the air	spread termination
spin spike in reference	momentum repair
"airborne" while lying	stricter metric
L2 task · this clip (backflip)	
hips extended, no tuck	tuck reward / limit
legs twisted (lotus)	hand-written term.
box height wrong (ext.)	scene repair

promote

fix verified on ≥ 2 tasks → promoted up
new failures enter L2 · stop criteria are layered the same way

adds entry

Next skill k+1
cartwheel · side flip · kung-fu kick
+ box backflip · wall flip (ext.)
inherits L0 + its family's L1 priors
→ fewer rounds, fewer interventions

(d) What changes vs. conventional video-to-humanoid pipelines

= scene-interaction extension (same harness)

Conventional
feed-forward,
tuned by hand

Video → Reconstruct → Retarget → Filter infeasible (✗ hard clips dropped) → Track (fixed rewards) → Policy

human tunes rewards per skill

Ours
harness-driven,
self-improving,
one laptop GPU

Video → Reconstruct (+ VLM labels · scene) → Retarget → **Repair reference** (physics priors · scene) → **Track** (library-driven edits) → Policy (→ G1 (real limits))

data repair

measure → LLM decides edits